

KHALED BEDDA

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Ontario, Canada

PROFESSIONAL EXPERIENCE

4 mos

Rivercity Innovations (Full-time) Saskatoon, Sask, Canada
Data Scientist May 2025 - Present

- Lead the AI and Data Science projects for cold-chain monitoring.
1. Built an end-to-end framework for temperature anomaly detection.
 2. Built a change point detection for sensor behavior analysis.
 3. Built a forecasting framework for sensor data breach (Azure).
 4. Proposed strategic plans to the C-suite executives.

1 yr 8 mos

Canada Revenue Agency (Full-time) Ottawa, ON, Canada
AI Tools & Infrastructure Engineer Aug 2023 - Mar 2025

1. Primary support for ArcGIS/MSSQL and R/Rstudio, Secondary for Anaconda
Project management, configuration, troubleshooting, and package certification.
2. Collaborate on AI Guidelines and GenAI Tools.
Review AI guidelines and strategy/ Participate in framework establishment and options analysis, contribute to the vision and development of Agency's GenAI tools
3. Conduct software testing and tools documentation.
Conduct software tools test cases and provide reports. Develop and maintain tools documentation including best practices
4. Collaborate on Presentation and Education-driven Events.
Presented AI concepts to a non-technical audience (Over 230 attendees)

1 yr 5 mos

Thunder Bay Regional Health Research Institute & LU (Full-time) Thunder Bay, ON, Canada
Graduate Research Assistant Sep 2021 - Jan 2023

1. Working on Distributed AI for 5G Networks - Thesis, Supervisor: [Dr. Zubair Md Fadlullah](#)
2. Working on **IoT** for healthcare projects - Lab projects
3. Conduct outdoor experiments as part of Lab research.

1 yr 8 mos

Lakehead University (Part-time, 13.5h/week) Thunder Bay, ON, Canada
Graduate Assistant Sep 2021 - April 2023

1. Assist in marking assignments, conducting sessions, and supervising labs.
2. Delivered a technical lecture on AWS SageMaker to 50+ students
 - Developed hands-on lab materials with SageMaker notebooks.
 - Assisted in curriculum design, integrating AWS services into coursework to align with industry trends.

SKILLS

Programming & MLOps: R, Python, C, C++, CUDA, MATLAB, TensorFlow, MLflow, Docker

Data Visualization & Analysis: NumPy, Pandas, Matplotlib, Seaborn, Plotly, SPSS, PowerBI, SAS.

Databases & GIS: SQL, MSSQL, MongoDB, ArcGIS, and QGIS.

Cloud Platforms: AWS (SageMaker(model training/pipelines), EC2, S3, Lambda, etc.), Azure

Web Development: Flask, Node.js, HTML5, CSS

Documentation & Management: LaTeX, Markdown, SharePoint, Git.

EDUCATION

Lakehead University

Sep 2021 - May 2023

M.Sc. in Computer Science - AI Specialization (Thesis based - Fully Funded)

Grade: A

Thesis: [New Paradigms of Distributed AI for Improving 5G-Based Network Systems Performance](#)

Cairo University

Sep 2016 - July 2021

B.Sc. (Hons) in Biomedical Engineering & Systems.

Grade: 87.6%

PUBLICATIONS

- K. Bedda, M. M. Fouda and Z. Md Fadlullah, "PC-SSL: Peer-Coordinated Sequential Split Learning for Intelligent Traffic Analysis in mmWave 5G Networks," 2023 IEEE 34th Annual International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC), Toronto, ON, Canada, 2023, pp. 1-6, doi: [10.1109/PIMRC56721.2023.10293872](#).
- G. Gad, A. Farrag, A. Aboulfotouh, K. Bedda, Z. M. Fadlullah and M. M. Fouda, "Joint Self-Organizing Maps and Knowledge-Distillation-Based Communication-Efficient Federated Learning for Resource-Constrained UAV-IoT Systems," in IEEE Internet of Things Journal, vol. 11, no. 9, pp. 15504-15522, 1 May1, 2024, doi: [10.1109/JIOT.2023.3349295](#).
- K. Bedda, Z. M. Fadlullah and M. M. Fouda, "Efficient Wireless Network Slicing in 5G Networks: An Asynchronous Federated Learning Approach," 2022 IEEE International Conference on Internet of Things and Intelligence Systems (IoTaIS), BALI, Indonesia, 2022, pp. 285-289, doi: [10.1109/IoTaIS56727.2022.9976007](#).
- A. Gamal, K. Bedda, N. Ashraf, S. Ayman, M. AbdAllah and M. A. Rushdi, "Brain Tumor Segmentation using 3D U-Net with Hyperparameter Optimization," 2021 3rd Novel Intelligent and Leading Emerging Sciences Conference (NILES), 2021, pp. 269-272, doi: [10.1109/NILES53778.2021.9600556](#).

PROJECTS

Generative Data Augmentation Approaches for EEG Abnormality Classification

Solved the issue of data imbalance via VAE data augmentation technique

A comparative analysis for permanent magnet synchronous motors Temperature prediction

Implemented an ML pipeline with regression based, tree based and RNN based models

Functional Data Analysis approach for High-dimensional Genetic Data Analysis

Solved the issue of high dimensionality and high covariance data by applying a functional data approach.

BS Graduation Project

Grade: Distinction

Brain Tumor Segmentation using 3D U-Net with Hyperparameter Optimization.

Autonomous Car

Using mobile camera, esp and a flask server to make a car that detects lanes and objects and drive autonomously.



Temperature controller Module (Proteus Simulation)

With Atmega32 Microcontroller I designed a System to provide a temperature-controlled functionality.



Indoor Localization System

Locate the esp in a building through WiFi signals and visualize on mobile app.

Digital Realtime Filter

Real time response filter simulation with 8051 microcontroller.

Progressive JPEG decoder

Python PyQt5 desktop app that decodes progressive jpeg images.

3D Robotic Body Model

Using OpenGL Library to create a model that simulates body joints movement with animation and interaction with objects.

Machine learning models for HCV data of Egyptians

Using machine learning models to predicts the phase of hcv for patients.



Huffman compression for images

C++ Desktop app that compress and decompress pgm images.

Signal Acquisition and Processing of EMG

Circuit for acquisition, filtering, and monitoring of EMG signals



Mathematical Modeling of Autoimmune process

A replicated research that models the autoimmune processes in body using differential equations.

CERTIFICATION

Architecting on AWS

Training was delivered over multiple informational sessions

Dec 2023

LANGUAGES

- English (IELTS/ 2023-2025) : Overall: 7 (R: 7, L: 7.5, S: 7, W: 6.5)
- English (TOEFL/ 2020-2022) : Overall: 101 (R: 29, L: 25, S: 23, W: 24)

REFERENCES

Hadi Amine

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